

Modeling the optimal seasonal monoclonal antibody administration strategy for respiratory syncytial virus (RSV) prevention based on age-season specific hospitalization rate of RSV in Suzhou, China, 2016–2022

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ABSTRACT

Background: The approval of nirsevimab brings light to reducing the heavy disease burden caused by respiratory syncytial virus (RSV). Considering the seasonality of RSV, the timing of administering monoclonal antibody (mAb) is critical to maximize health utility. This study aimed to model and seek the optimal seasonal mAb administration strategy for preventing RSV-associated hospitalization.

Methods: Age-season specific hospitalization rates for RSV-associated acute lower respiratory infection (RSV-ALRI) were estimated from a hospital-based birth cohort. Using these rates, we simulated and evaluated the effect of diverse mAb administration strategies on preventing RSV-ALRI hospitalization. Optimal strategies were selected based on their effectiveness and relative cost-effectiveness.

Results: Compared with the year-round strategy of administration mAb at birth for all children, 291 out of the 854 candidate strategies, featuring diverse administration timing and age thresholds, demonstrated a greater number of averted RSV-ALRI hospitalizations and a lower number needed to treat (NNT). The NNT represents the number of mAb doses needed to prevent one case of RSV-ALRI hospitalization. Among the 291 strategies, administration mAb to children born in July–January or August–January at birth and administering to the remaining <12 months old children in September, exhibited the highest increase in averted RSV-ALRI hospitalizations than the year-round strategy, with a magnitude of 23 %, while also achieve an 18 % reduction in NNT.

Conclusion: Administering monoclonal antibodies to children born in July to January at birth, and administering to the remaining <1-year-old children in September or October would be the optimal seasonal mAb administration strategy for children in Suzhou, China.

1. Introduction

Respiratory syncytial virus (RSV) is the most common viral pathogen causing acute lower respiratory infection (ALRI) among young children globally [1]. In 2019, RSV caused an estimated 33 million episodes, 3.6 million hospitalizations, and 26,300 in-hospital deaths worldwide in children aged < 5 years [2]. Recent research has suggested a potential association between respiratory syncytial virus infection during infancy and childhood asthma and recurrent wheezing while establishing a

causal relationship remains inconclusive [3,4]. Despite the uncertainty regarding its long-term impact, given the substantial disease burden associated with RSV in pediatric populations, preventing RSV infection holds a significant public health priority.

The RSV prevention measures currently marketed and in various stages of clinical trials encompass maternal vaccines, pediatric vaccines, and pediatric monoclonal antibodies (mAb) [5]. Preceding the approval of nirsevimab, palivizumab is the only prophylaxis for RSV, but its use was restricted to high-risk population for prevent severe infection [6,7].

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Recently, Nirsevimab, a single-dose long-acting antibody targeting RSV pre-fusion F protein has been proven to have 62 % efficacy in preventing RSV-associated lower respiratory tract hospitalization [8,9]. In November 2022 and July 2023, nirsevimab obtained regulatory approval for preventing RSV infection in infants in Europe and the United States, respectively [10,11], which makes it the first preventive option to safeguard the broad newborn infant population from RSV infection. As passive immunization interventions, monoclonal antibodies provide immediate protective effects upon administration but confer protection for a limited duration. Considering the seasonal pattern of RSV infection within temperate regions, investigating the optimal timing for monoclonal antibody administration is important to enhance preventive strategies' effectiveness. Nonetheless, research in this field remains limited [12]. From both an effectiveness and relative cost-effectiveness perspective, this study aimed to model and seek the optimal administration timing and age threshold for implementing a seasonal mAb administration strategy to preventing RSV-associated hospitalization.

In this study, we first estimated the age-season specific hospitalization rate of RSV-associated ALRI (RSV-ALRI) based on a hospital-based birth cohort. With the age-season specific hospitalization rates estimate as a crucial parameter, we subsequently conducted simulations to evaluate the effect of diverse RSV seasonal mAb administration strategies on preventing RSV-ALRI hospitalization. The candidate strategies are divided into four categories, comprising 855 in total, each with a distinct administration timing or age threshold. Finally, by striking a balance between effectiveness and cost-effectiveness, our study aims to identify the optimal seasonal mAb administration strategy from the candidate strategies, thus providing evidence-based support for decision-making.

2. Methods

2.1. Study site

The study was conducted at Soochow University Affiliated Children's Hospital (SCH) in Suzhou City of Jiangsu Province, southeast China. SCH is a tertiary Grade-A pediatric hospital with two campuses, Jingde Road Campus and Industrial Park Campus, and serves approximately 65 % of the population under age 5 years in downtown Suzhou. There are about 1.9 million outpatient visits (including emergency visits) and 45,000 hospitalizations at SCH annually [13]. At SCH, all medical records, including laboratory testing results, are digitized and saved by Hospitalization Information System (HIS) for multiple years. International Classification of Diseases 10th Revision (ICD-10) coding is used for discharge diagnoses with coding performed by the treating physician.

2.2. Hospital-based birth cohort

The birth cohort is a dynamic hospital-based cohort [13]. From December 12, 2016 onwards, any child born on or after this date and seeking medical care, encompassing both outpatient and inpatient services at SCH was automatically enrolled as a participant in the cohort. All children were followed through the SCH HIS database until December 31, 2022. The SCH HIS database contained patient demographics, visiting or admission ID, date of visit or admission, all discharge diagnosis ICD-10 codes, disease outcome, death date if applicable, and laboratory data (including pathogen test). RSV testing is routinely conducted for all hospitalized ALRI cases, utilizing direct immunofluorescence assay (DFA) before 2022 and transitioning to reverse transcription-polymerase chain reaction (RT-PCR) thereafter in 2022. By linking the name, sex, and birth date of the cohort subjects with the vaccine registry database of the Suzhou Center for Disease Control and Prevention (CDC), we retrieve the vaccination records of partial cohort subjects to identify their residence. The database linking process

was conducted twice, once in September 2019 and again in November 2021, for all cohort subjects born between December 12, 2016, and December 31, 2018. Data analysis was conducted on the cohort subjects born between December 12, 2016, and December 31, 2018 and residing in Suzhou city.

2.3. Incidence density estimation

ALRI episodes were identified by referencing hospital discharge ICD-10 diagnosis codes within the range of J09-J18 (influenza and pneumonia) or J20-J22 (other acute lower respiratory infections). RSV-ALRI cases were defined as ALRI cases that tested positive for RSV. ALRI-related outpatient visits or hospitalizations occurring within a 30-day interval were recorded as a single episode. The incidence density of ALRI episodes, ALRI hospitalizations, and RSV-ALRI hospitalizations were calculated overall, as well as stratified by birth year and age group. The incidence density of RSV-ALRI hospitalization among children under 2 years old was meticulously calculated, stratified by month of birth and age in months, to serve as crucial disease burden parameters for subsequent modeling analysis.

2.4. Strategy evaluation

Given the seasonality of RSV and the limited protection duration of mAb, researchers proposed that for infants born outside the RSV season, mAb can be administered just before the start of the season, whereas infants born during the RSV season can be immunized at birth [14,15]. Similarly, the candidate strategies in this study are divided into four categories:

- Year-round administration at birth strategy (S1), administering mAb at birth regardless of their birth month;
- Seasonal administration at birth strategy (S2.s.d), administering mAb at birth only to children born within month s and the subsequent $d-1$ months (a total of d months birth window), where s ranges from 6 to 12, and d ranges from 4 to 11;
- Administration at season strategy (S3.m.a), administering mAb to children age less or equal than a month old in month m every year, where m ranges from 6 to 12, and a takes a value of either 6 or 12;
- Seasonal with annual catch-up strategy (S4.s.d.m.a), a combination of S2.s.d and S3.m.a, administering mAb at birth to children born within month s and the subsequent $d-1$ months, and administering mAb to children age less or equal than a month old in month m for children born outside above d months birth window.

Guided by the directed acyclic graph (DAG) depicted in Fig. 1A, we evaluated the effects and relative cost-effectiveness of different RSV immunization strategies for preventing RSV-associated ALRI hospitalizations in children under 2 years old. According to the back-door criteria [16,17], age and season need to be adjusted when evaluating the effect of immunization strategy on RSV-ALRI hospitalization. During the strategy evaluation process, we compute the values of process variables, stratified by birth month and age in month, and then aggregate the results to determine the overall effectiveness of a specific strategy. (Fig. 1B) A hypothetical cohort of 12 million population, with a birth rate of 7 ‰, was simulated for ease of numerical calculation. Referring to the clinical trial results of nirsevimab, mAb is assumed to have an efficacy of 60 % and provide protection for 5 months [8,9]. Based on the influenza vaccine coverage rates in Suzhou city, the mAb's coverage rate is assumed to be 20 %. The age-season specific hospitalization rate of RSV-ALRI is supposed to conform to Poisson distributions with means (λ) estimated from the aforementioned birth cohort. The number of averted RSV-ALRI hospitalizations for different strategies was calculated by aggregating the stratified number of averted RSV hospitalizations based on month of birth and age in months. The total number of mAb doses consumed serves as an indicator of cost for different strategies.

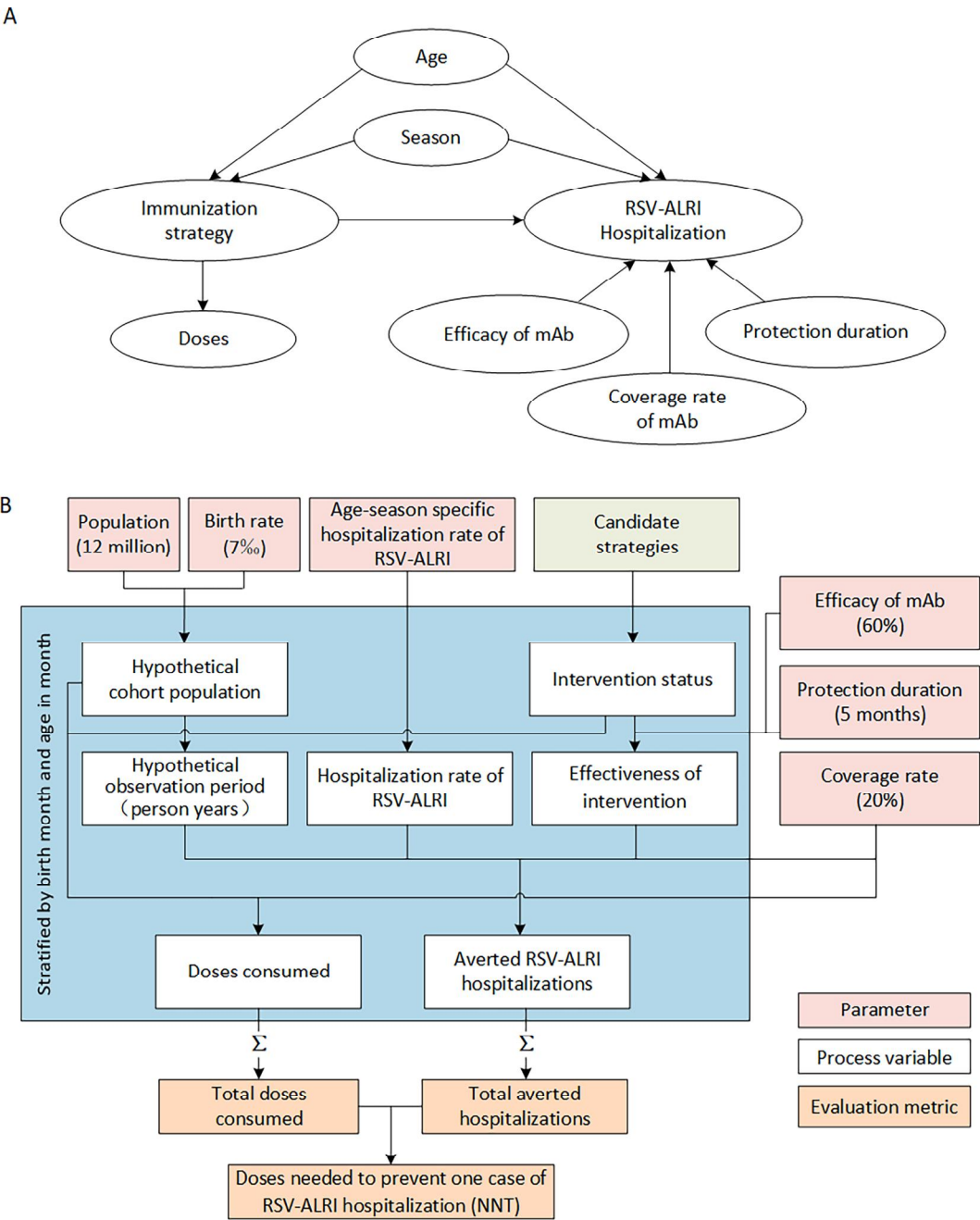


Fig. 1. Directed acyclic graph and model structure for strategy evaluation. RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection; mAb: monoclonal antibody; Number in parentheses represents the default value of the model parameter.

The number needed to treat (NNT), which represents the number of mAb doses needed to prevent one case of RSV-ALRI hospitalization, is used to assess the relative cost-effectiveness of different strategies. In each scenario, we performed 100 simulation iterations and presented the mean of the results for reporting and strategy comparison. (Table 1 [18]).

2.5. Sensitivity analysis and software

In sensitivity analysis, we tune the parameters encompassing mAb efficacy (50 % and 70 %), duration of protection(3,4,6,7 months), and coverage rate (40 % and 60 %) to explore their influence on the outcomes. All analysis were performed using R software(version 4.2.0) with code available at Github [19]. An online Shiny application for this paper can be accessed at <https://slr95.shinyapps.io/rsv-shiny>.

2.6. Ethical approvalxxx

This study was approved by the School of Public Health, Fudan University Human Research Ethics Committee and the Soochow University Affiliated Children Hospital's Human Research Ethics Committee.

3. Results

From December 12, 2016 to December 31, 2022, a total of 719,584 subjects were identified from 4,996,565 medical records of SCH. 334,988 subjects born between December 12, 2016 and December 31, 2018 were attempted to link with the CDC vaccine registry database. Among them, 173,052(51.7 %) subjects were successfully matched and considered to reside in Suzhou City, thus included in the final analysis

Table 1
Summary of key input parameters, process variables, and evaluation metrics for strategy evaluation.

Parameters or variables	Value	Source
<i>Input Parameters</i>		
Population	12 million (default)	Referring to the permanent population of Suzhou City [18]
Birth Rate	7 ‰ (default)	Referring to the birth rate of the registered population in Suzhou [18]
Hospitalization Rate of RSV-ALRI for children born in month <i>i</i> during the <i>j</i> month of age(<i>HR_{ij}</i>)	Poisson distribution with a parameter of λ , where λ is estimated from the hospital-based birth cohort	Specially, the RSV-ALRI hospitalization rate for children in the first month after birth is assumed to be equal to the RSV-ALRI hospitalization rate in the second month, namely $HR_{i0} = HR_{i1}$
mAb Efficacy	60 % (default) 50%,70 % (for sensitive analysis)	Referring to the clinical trial result of nirsevimab [8,9]
Protection duration (months)	5 (default) 3,4,6,7 (for sensitive analysis)	Referring to the clinical trial result of nirsevimab [8,9]
Coverage Rate(Cov)	20% (default) 40%,60% (for sensitive analysis)	Referring to the influenza vaccine coverage rates in Suzhou city
<i>Process Variables</i>		
Population born in month <i>i</i> (<i>Pop_i</i>)	Population × Birth Rate × (1/12)	<i>i</i> range from 1 to 12
Person-years contributed by children born in month <i>i</i> during the <i>j</i> month of age (<i>PY_{ij}</i>)	<i>Pop_i</i> × (1/12)	<i>j</i> range from 0 to 23
Intervention status for children born in month <i>i</i> during the <i>j</i> th month of age (<i>Int_{ij}</i>)	0 for no intervention, 1 for intervention	Calculated based on diverse immunization strategies
mAb effectiveness for children born in month <i>i</i> during the <i>j</i> th month of age (<i>ME_{ij}</i>)	0 or mAb efficacy	Calculated based on intervention status, protection duration, and mAb efficacy
<i>Evaluation metrics</i>		
Total number of averted RSV-ALRI hospitalizations (Averted Hospitalizations)	$\sum_{i=1}^{12} \sum_{j=0}^{23} PY_{ij} HR_{ij} ME_{ij} Cov$	
Total number of doses consumed(Doses)	$\sum_{i=1}^{12} \sum_{j=0}^{23} Pop_i Int_{ij} Cov$	
The number of mAb doses needed to prevent one case of RSV-ALRI hospitalization(NNT)	Doses / Averted hospitalizations	

RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection, mAb: monoclonal antibody.

(Fig. 2).

3.1. Incidence density

During the observation period from 0 to 5 years of age, a total of 97,786 ALRI episodes were observed among the study subjects. Out of these, 17,203 (17.6 %) cases required hospitalization due to the severity of the condition. Among the hospitalized ALRI cases, 11,931 (69.4 %) underwent RSV testing, with 2,021 (16.9 %) cases testing positive for RSV. The incidence density of ALRI episodes, ALRI hospitalization, and

RSV-ALRI hospitalization for children under 5 years old in Suzhou, China were 117.6 (95 % CI: 116.9–118.3), 20.7 (95 % CI: 20.4–21.0), and 3.5 (95 % CI: 3.4–3.6) per 1000 person-years, respectively. The incidence density of RSV-ALRI hospitalization was highest among children aged 0–5 months, with a rate of 13.9 (95 % CI: 13.1–14.7) per 1000 person-years. The incidence rates gradually decreased with increasing age, with rates of 5.7 (95 % CI: 5.2–6.2), 2.6 (95 % CI: 2.3–2.8), 1.4 (95 % CI: 1.3–1.6), and 1.1 (95 % CI: 1.0–1.2) per 1000 person-years observed among children aged 6–11, 12–23, 24–35, and 24–59 months, respectively. By stratifying the cohort subjects based on their birth year, the incidence densities were calculated within each stratum. (Table 2).

3.2. Age-season specific RSV-ALRI hospitalization rate

The study subjects were stratified by month of birth for analysis, and the incidence density of ALRI hospitalization, namely the hospitalization rate of RSV-ALRI, for children born in different months is presented in Fig. 3 and supplementary materials. The hospitalization rate of RSV-ALRI varies significantly among children born in different months within the same age group before 2 years old. Children born between July and November exhibit the highest hospitalization rate of RSV-ALRI during the 0–5 months age period, followed by a sharp decrease after 6 months of age. Children born in December, January, May, and June exhibit a higher hospitalization rate of RSV-ALRI during the 0–11 months age periods. Children born in February to April exhibit a lower hospitalization rate of RSV-ALRI during the 0–5 months age period but experience a peak hospitalization rate during the 6–11 months age period.

We further calculated the monthly hospitalization rate of RSV-ALRI among children under 2 years old, stratified by month of birth, as shown in Fig. 4. On one hand, the hospitalization rate of RSV-ALRI shows fluctuations among children born in the same month due to aging and seasonal variations in RSV activity. On the other hand, notable variations in the hospitalization rate of RSV-ALRI are observed among children born in different months at the same age. The combined influence of age and seasonal factors leads to significant heterogeneity in the hospitalization rate of RSV-ALRI, which also highlights the necessity of stratifying based on these two factors in strategy evaluation.

3.3. Strategy evaluation

Under the default parameter setting, the year-round strategy (S1) consumes an average of 16,800 doses of monoclonal antibodies, preventing 84(standard deviation,SD:2) cases of RSV-ALRI hospitalization, with an NNT of 199(SD:6) on average. Among the other 854 candidate strategies, 291 strategies are deemed superior to S1, as they exhibit a lower NNT and a greater number of averted hospitalizations; 170 strategies are comparable to S1, as they exhibit a lower NNT but fewer averted hospitalizations; and 393 strategies are deemed inferior to S1, as they exhibit a higher NNT and fewer averted hospitalizations. (Figure S1 in supplementary material) The 291 superior strategies are exclusively seasonal with annual catch-up strategies.

Fig. 5 illustrates the NNT and averted RSV-ALRI hospitalizations for the 291 superior strategies. When the age threshold for the season-with catch up strategy is set at 6 months, the relationship between the NNT and total averted hospitalizations follows a roughly inverted U-shape. Strategy with the lowest NNT demonstrate limited total averted hospitalizations, while the strategy achieving the highest averted hospitalizations typically does not correspond to the smallest NNT. When the age threshold for the season-with catch up strategy is set at 12 months, the relationship between the NNT and total averted hospitalizations exhibits an approximate negative linear trend. As the total averted hospitalizations increases, the NNT for each strategy decreases, given that the total mAb doses consumed remain constant across all strategies in this case. Overall, under identical birth month window for at-birth administration

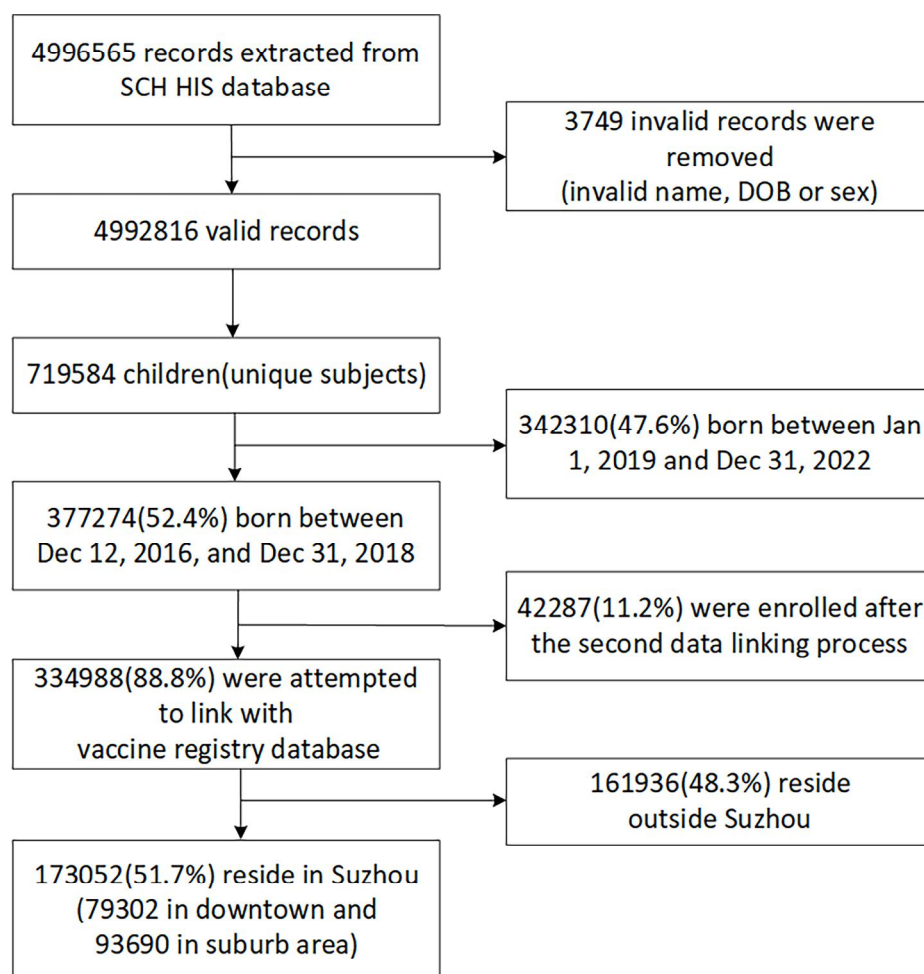


Fig. 2. Subject inclusion flowchart for final analysis. SCH: Soochow university affiliated children's hospital; HIS: hospital information system; DOB: date of birth.

and catch-up timing condition, setting the catch-up age threshold at 12 months leads to more averted RSV-ALRI hospitalizations than the 6-month threshold, albeit with a slight increase in NNT. In terms of the selection of birth month window for at-birth administration, under identical catch-up timing and age threshold condition, the birth month window for strategies with smaller NNT predominantly falls within the range of June to January. In terms of the selection of catch-up timing, under identical birth month window for at-birth administration condition, the catch-up timing for strategies with smaller NNT predominantly falls within August to October.

The top 20 superior strategies with the lowest NNT and the top 20 superior strategies with the highest number of averted hospitalizations are depicted in Fig. 6. Compared to the year-round strategy, strategies S4.7.6.10.6 and S4.7.7.11.6 demonstrated the highest reduction in average NNT, with a decrease of 34 %, while also achieving a 1 % total increase in averted hospitalizations; strategies S4.7.7.9.12 and S4.8.6.9.12 exhibited the highest increase in average averted hospitalizations, with a magnitude of 23 %, while also achieve an 18 % reduction in NNT.

The sensitivity analysis results demonstrate that the relative rankings of candidate strategies compared to the year-round strategy are not sensitive to changes in mAb efficacy and coverage rate. Across mAb efficacy values of 0.5 and 0.7, and coverage rates of 0.4 and 0.6, the top 20 strategies identified align closely with those selected under the default parameter setting. When the assumed mAb protection duration varies between 3 and 7 months, the rankings of the top 20 superior strategies exhibit minor shifts. Compared to strategies S4.7.7.9.12 and S4.8.6.9.12, strategies S4.7.7.10.12 and S4.8.6.10.12 demonstrate slight

advantages in both effectiveness and efficiency when the mAb protection duration is reduced to 4 or 3 months. Conversely, strategies S4.7.7.8.12 and S4.8.6.8.12 demonstrate slight advantages in both effectiveness and efficiency when the mAb protection duration is increased to 6 or 7 months. (Figure S2-S9 in supplementary material).

4. Discussion

In this study, we estimated the incidence density of RSV-ALRI hospitalization among children under 5 years old, which has been rarely reported in previous research from China. The incidence density of RSV-ALRI hospitalizations estimated in this study for children under 6 months, 1 year, and 5 years is 13.9, 9.8, and 3.5 per 1,000 person-years, respectively. These results are consistent with the rates reported in the United States[20], which were 17 per 1000 for children under 6 months and 3 per 1000 for children under 5 years of age, and comparable with those in Europe[21], which were 18 per 1000 for children under 1 year old. Considering the influence of COVID-19 lockdowns on patient healthcare-seeking behavior and potential disruptions in hospitalization rate estimates, we conducted stratified estimations based on birth year. In theory, the implementation of COVID-19 lockdown measures, resulting in reduced healthcare-seeking behavior, could decrease disease incidence density during the pandemic. Consequently, children born in 2018, within the 12–23 months age range (2019–2020), are expected to exhibit a lower hospitalization rate of RSV-ALRI compared to those born in 2016–2017 within the same age range (2018–2019). In fact, children born in 2018 demonstrated a hospitalization rate of 2.5 (95 % CI: 2.2–2.9) per 1000 person-years at ages 12–23 months, slightly

Table 2
Incidence density of ALRI episodes, ALRI hospitalization and RSV-ALRI hospitalization among children under 5 years old in Suzhou, China.

Age group	Birth year	Person years	ALRI episode	ALRI hospitalization	Hospitalized ALRI cases tested for RSV	Hospitalized ALRI cases test positive for RSV	Positive rate of RSV	Incidence density (per 1000 children-years)		
								ALRI episode	ALRI hospitalization	RSV-ALRI hospitalization
0–5 m	2016–2017	50,061	8415	2086	1669 (80.0)	487	29.2	168.1 (164.5–171.7)	41.7 (39.9–43.5)	12.2 (11.2–13.2)
	2018	36,465	7236	2197	1851 (84.3)	498	26.9	198.4 (193.9–203.1)	60.2 (57.8–62.8)	16.2 (14.9–17.6)
	Subtotal	86,526	15,651	4283	3520 (82.2)	985	28.0	180.9 (178.1–183.7)	49.5 (48.0–51.0)	13.9 (13.1–14.7)
6–11 m	2016–2017	50,061	9085	1286	1025 (79.7)	191	18.6	181.5 (177.8–185.2)	25.7 (24.3–27.1)	4.8 (4.2–5.4)
	2018	36,465	9069	1400	1110 (79.3)	201	18.1	248.7 (243.6–253.9)	38.4 (36.4–40.5)	7.0 (6.1–7.9)
	Subtotal	86,526	18,154	2686	2135 (79.5)	392	18.4	209.8 (206.8–212.9)	31.0 (29.9–32.2)	5.7 (5.2–6.2)
12–23 m	2016–2017	100,122	18,196	2279	1727 (75.8)	197	11.4	181.7 (179.1–184.4)	22.8 (21.8–23.7)	2.6 (2.3–2.9)
	2018	72,930	8799	1669	1161 (69.6)	127	10.9	120.6 (118.1–123.2)	22.9 (21.8–24.0)	2.5 (2.2–2.9)
	Subtotal	173,052	26,995	3948	2888 (73.2)	324	11.2	156.0 (154.1–157.9)	22.8 (22.1–23.5)	2.6 (2.3–2.8)
24–35 m	2016–2017	100,122	8143	1323	902 (68.2)	79	8.8	81.3 (79.6–83.1)	13.2 (12.5–13.9)	1.2 (1.0–1.4)
	2018	72,930	4834	1098	620 (56.5)	78	12.6	66.3 (64.4–68.2)	15.1 (14.2–16.0)	1.9 (1.6–2.2)
	Subtotal	173,052	12,977	2421	1522 (62.9)	157	10.3	75.0 (73.7–76.3)	14.0 (13.4–14.6)	1.4 (1.3–1.6)
36–59 m	2016–2017	200,244	13,777	2357	1142 (48.5)	99	8.7	68.8 (67.7–70.0)	11.8 (11.3–12.3)	1.0 (0.9–1.2)
	2018	112,109	10,232	1508	724 (48.0)	64	8.8	91.3 (89.5–93.1)	13.5 (12.8–14.1)	1.2 (1.0–1.4)
	Subtotal	312,353	24,009	3865	1866 (48.3)	163	8.7	76.9 (75.9–77.8)	12.4 (12.0–12.8)	1.1 (1.0–1.2)
0–11 m	2016–2017	100,122	17,500	3372	2694 (79.9)	678	25.2	174.8 (172.2–177.4)	33.7 (32.6–34.8)	8.5 (7.9–9.1)
	2018	72,930	16,305	3597	2961 (82.3)	699	23.6	223.6 (220.2–227.0)	49.3 (47.7–51.0)	11.6 (10.9–12.5)
	Subtotal	173,052	33,805	6969	5655 (81.1)	1377	24.4	195.3 (193.3–197.4)	40.3 (39.3–41.2)	9.8 (9.3–10.3)
0–23 m	2016–2017	200,244	35,696	5651	4421 (78.2)	875	19.8	178.3 (176.4–180.1)	28.2 (27.5–29.0)	5.6 (5.3–5.9)
	2018	145,860	25,104	5266	4122 (78.3)	826	20.0	172.1 (170.0–174.3)	36.1 (35.1–37.1)	7.2 (6.8–7.7)
	Subtotal	346,104	60,800	10,917	8543 (78.3)	1701	19.9	175.7 (174.3–177.1)	31.5 (31.0–32.1)	6.3 (6.0–6.6)
0–59 m	2016–2017	500,610	57,616	9331	6465 (69.3)	1053	16.3	115.1 (114.2–116.0)	18.6 (18.3–19.0)	3.0 (2.9–3.2)
	2018	330,899	40,170	7872	5466 (69.4)	968	17.7	121.4 (120.2–122.6)	23.8 (23.3–24.3)	4.2 (4.0–4.4)
	Subtotal	831,509	97,786	17,203	11,931 (69.4)	2021	16.9	117.6 (116.9–118.3)	20.7 (20.4–21.0)	3.5 (3.4–3.6)

ALRI: acute lower respiratory infection, RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection

lower than the hospitalization rate of 2.6 (2.3–2.8) per 1000 person-years observed for those born in 2016–2017. This minor difference indicates that the impact of the COVID-19 pandemic on hospitalization rate estimates for the studied cohort during the pre-2-year life stage is limited. From a temporal perspective, the majority of observational person-time under 2 years old in this study was free of COVID-19 impact, except for the cohort subject born in 2018 who experienced the COVID-19 outbreak during their 12–23 months age period. Due to the majority of the disease burden for RSV occurring before one year old, it can be deemed that the estimated age-season specific hospitalization rates of RSV-ALRI in this study for children under 2 years old are generally reliable.

RSV hospitalization rates vary significantly with age and season. The confluence of age and seasonal factors leads to disparate disease incidence risks among children born in different months, especially during the period before 2 years old. Children born in January and from June to December encounter the highest risk of RSV-ALRI hospitalization during the initial 0–5 months, indicating that immunization with monoclonal

antibodies at birth probably offers maximum protection for this sub-population. Conversely, for infants born from February to May who confront the highest RSV-ALRI hospitalization risk during the 6–11 months age period, an immunization at season strategy could effectively prevent a greater number of cases. The results of the strategy evaluation further support this conclusion. Notably, the period from birth to the first visit to SCH is left censored for the study subjects. Only 5,068 subjects had their first visit to SCH at an age younger than 1 year, with a median age of 18 days. Because of the limited observed person-time in the 0–1 month age range, the estimated hospitalization rate for the first month is underestimated in our study. (Fig. 4) During the strategy evaluation, we approximate the hospitalization rate for the 0–1 month age range with the rate for the adjacent age range (1–2 months) to expect more realistic results.

The results of strategy evaluation demonstrate that opting for appropriate seasonal with annual catch-up strategies, in contrast to the year-round strategy, not only achieves a notable reduction in NNT but also yields more averted hospitalizations. When the age threshold for the

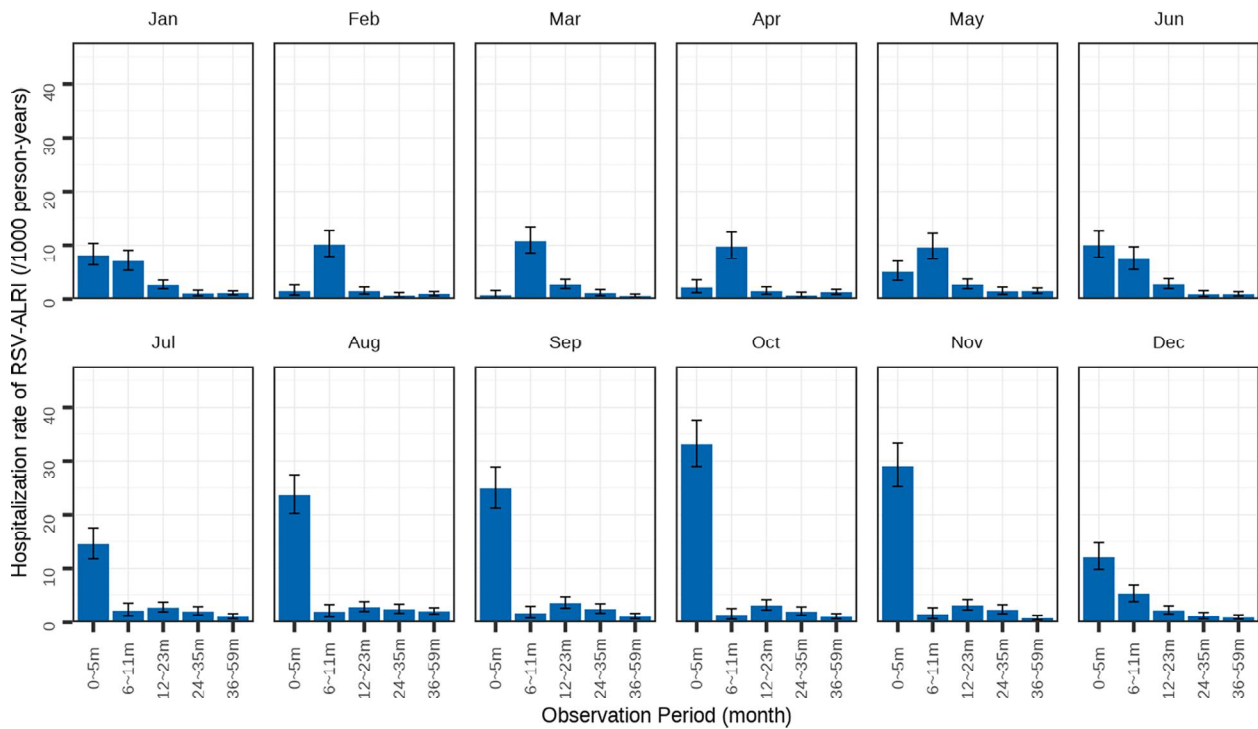


Fig. 3. Hospitalization rate of RSV-ALRI by birth month among children under 5 years old in Suzhou, China. RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection.

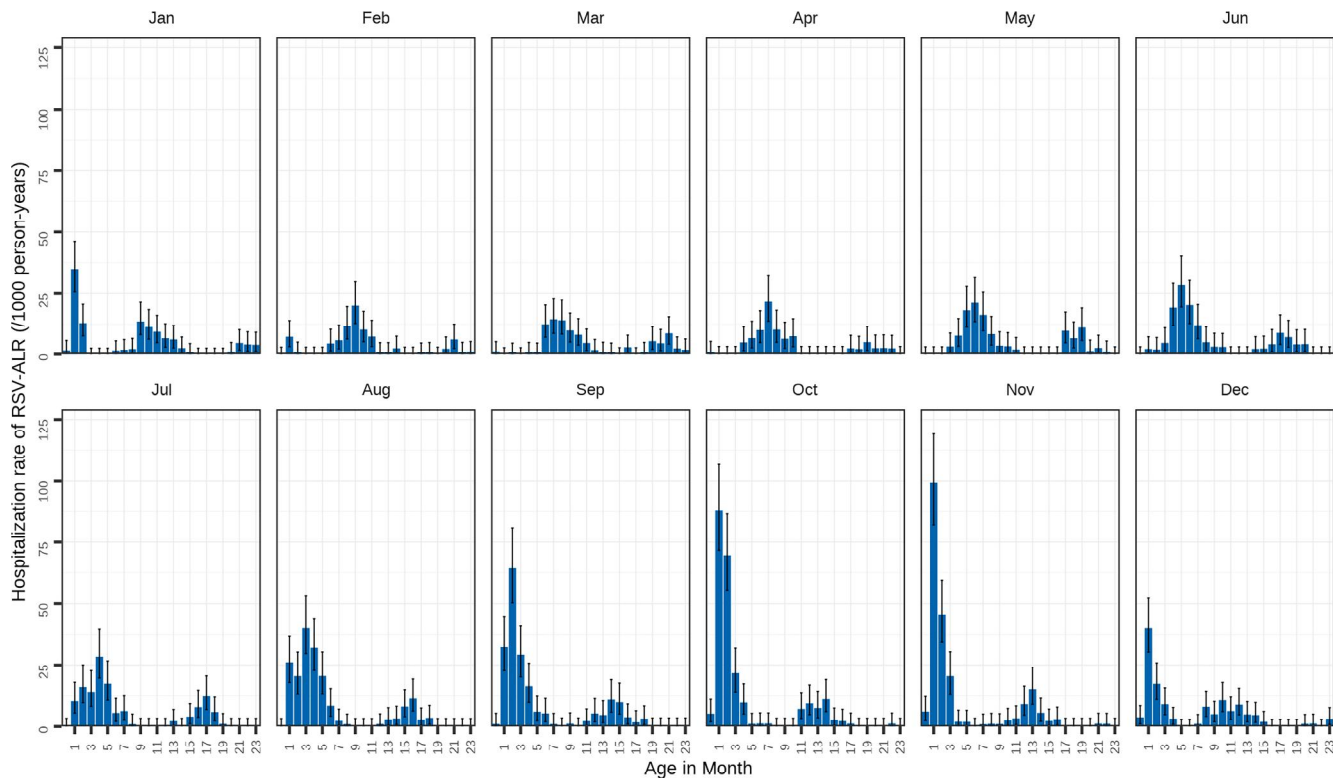


Fig. 4. Monthly hospitalization rate of RSV-ALRI by birth month among children under 2 years old in Suzhou, China. RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection.

season-with catch-up strategy is set at 6 months, strategies with the lowest NNT tend to target the birth month window with the highest RSV hospitalization rate, excluding populations with relatively lower risk. Consequently, the strategy with the lowest NNT does not yield a high

total of averted hospitalizations. The overall effectiveness and cost-effectiveness exhibit an inverted U-shaped pattern. When the age threshold for the season-with catch-up strategy is set at 12 months, different strategies consume the same monoclonal antibody doses.

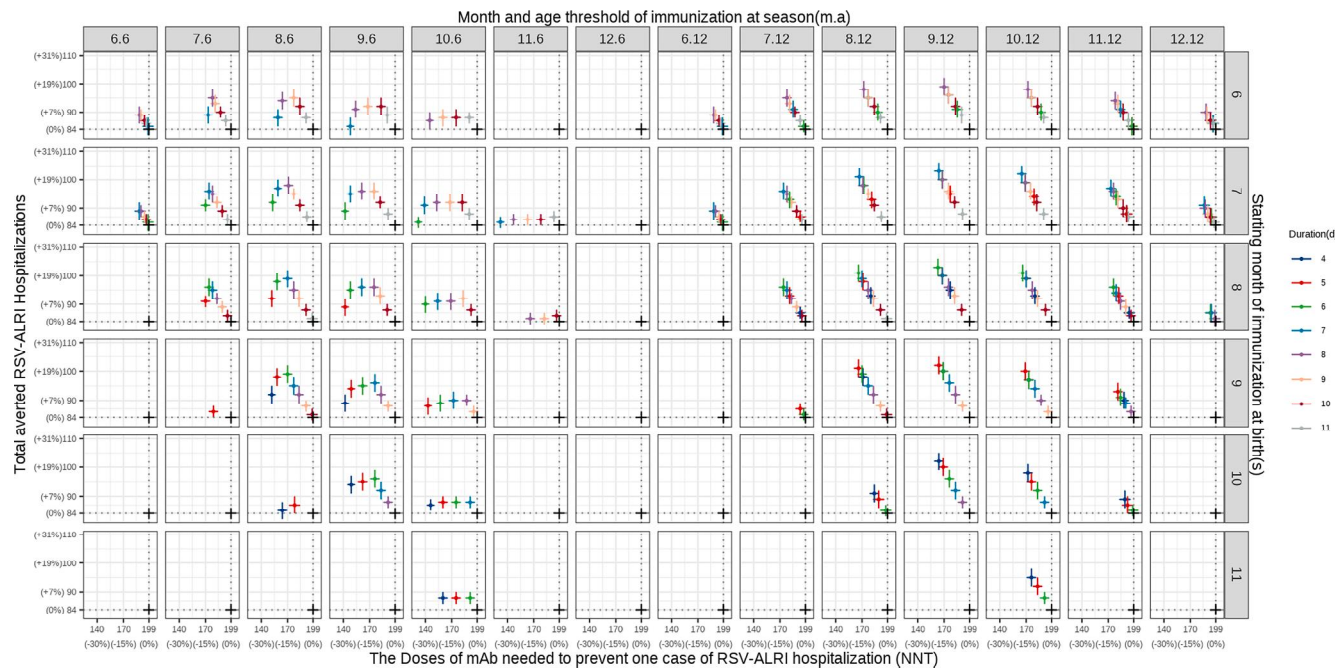


Fig. 5. The NNT and averted RSV-ALRI hospitalizations comparison for 309 superior strategies relative to the year-round administration at birth strategy. NNT: the dose of monoclonal antibody needed to prevent one case of RSV-ALRI hospitalization; RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection. Dashed lines represent the average NNT and averted hospitalization for the year-round administration at birth strategy (S1). Each point represents the average result of one superior seasonal with catch-up strategy(S4.s.d.m.a), which means administering monoclonal antibody at birth to children born within month s and the subsequent d-1 months and administering monoclonal antibody to children age less or equal than a month old in month m for children born outside above d months birth window. Error bars depict the standard deviation of NNT or averted RSV-ALRI hospitalizations.

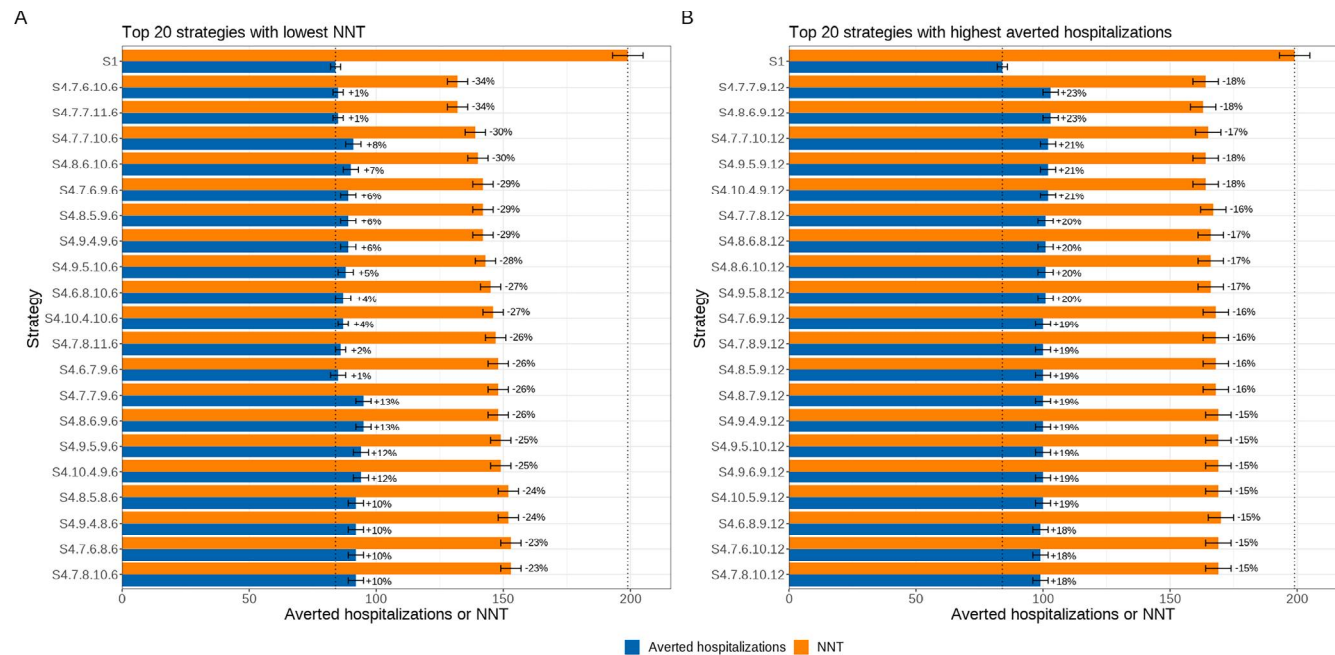


Fig. 6. Top 20 superior strategies with the lowest NNT (A) and highest averted hospitalizations (B). RSV-ALRI: respiratory syncytial virus associated acute lower respiratory infection. NNT: Dashed lines represent NNT (Number Needed to Treat) and averted hospitalization for the year-round strategy (S1). Error bars depict the standard deviation of NNT or averted hospitalizations. The percentage represents the decrease in average NNT or increase in averted hospitalizations compared with the year-round administration at birth strategy.

Consequently, the NNT decreases monotonically with an increase in averted hospitalizations. On the whole, the optimal period for administering mAb at birth is from July or August onwards, extending over a span of 6 or 7 months. The result is consistent with the trend of age-season specific hospitalization rate of RSV-ALRI. Infant born between

July and January face the highest risk of RSV hospitalization during the period up to six months of age (Fig. 4). Therefore, administering mAb at birth offers optimal protection against RSV hospitalization for children born between July and January. For infants who did not receive immunization at birth, mAb should be administered before the onset of the

RSV epidemic season. From a relative cost-effectiveness perspective, augmenting the above administration at birth strategy by administering mAb to remaining children under 6 months of age in October or November could reduce dose consumption by around 30 %, while maintaining equivalent or improved preventive effect. From the effect perspective, further expanding the threshold age of administration at season strategy to below 12 months, and administering immunization in September or October yields an approximately 20 % rise in the total averted hospitalizations relative to the year-round strategy, along with an 18 % reduction in mAb consumption. There is a trade-off between cost-effectiveness and overall effect. The most cost-effective strategies often exclude a portion of the population with relatively lower risk, which challenges the equity of healthcare resource distribution. Setting the upper threshold age of administration at season strategy at 6 months would enhance cost-effectiveness, but it would result in the exclusion of children who haven't received immunization at birth and are not older than 6 months when the epidemic season begins. In light of both fairness and overall effect, it is proposed to set the threshold age of administration at season strategy as 12 months and administering mAb in September or October annually, thereby ensuring that all children are covered by either administration at birth or at season strategy. Given that nirsevimab's clinical trial results were based on a 150-day observation period [9], this study assumed a default monoclonal antibody protection duration of 5 months. The actual protection duration of nirsevimab may surpass 5 months. Sensitivity analysis results indicated that if the protection duration extends to 6 or 7 months, it might be reasonable to consider advancing the annual catch-up timing to August or September.

On May 25, 2023, the Center for Drug Evaluation (CDE) of the China National Medical Products Administration (NMPA) accepted the Biologics License Application (BLA) for nirsevimab. The introduction of nirsevimab to the market in China may require some time. On August 3, 2023, nirsevimab was recommended by the United States Advisory Committee on Immunization Practices (ACIP) for infants aged < 8 months who are born during or entering their first RSV season and for infants and children aged 8–19 months who are at increased risk for severe RSV disease and are entering their second RSV season [22]. ACIP recommended that the optimal timing for nirsevimab administration is shortly before the RSV season begins. This study provides data on the effectiveness and cost-effectiveness of various administration strategies with differing administration timing or age threshold. In China, vaccines are categorized as either NIP (National Immunization Program) vaccines or non-NIP vaccine. The NIP vaccines are mandatorily administered and the cost are borne by government. The non-NIP vaccines are voluntarily administered and self-funded by citizens [23]. For this reason, the coverage rate of NIP vaccines is high (>80 %), while the coverage rate of non-NIP vaccines varies ranges from 1.8 % to 67.1 % [24]. Due to financial constraints and other reasons, many WHO recommended vaccines, such as influenza vaccine and pneumococcal conjugate vaccine (PCV), have not yet been included into NIP in China [23]. Nirsevimab is priced at \$495 per dose for 50 mg and 100 mg doses in United States [25]. Referencing this price, the probability of nirsevimab being included in the NIP after approval appears to be low. Therefore, in this study, the default coverage rate of nirsevimab is assumed to be 20 %, similar to the vaccination rate of non-NIP influenza vaccines in Suzhou, China. In this study, we assumed the coverage rate is unaffected by different strategies, and we compare the effect of different strategies under uniform coverage rate. However, the real-world situation may differ from this assumption. For example, mAb administration at birth can be implemented within the birth hospital, while the annual catch-up administration may rely on guardians actively visiting vaccination clinics. The variability in accessibility could result in different coverage rates across diverse strategies. Due to the difficulty in quantifying this variability, this study did not account for variations in coverage rates among different strategies, constituting one of the limitations of this research. Apart from the primary cost-effectiveness factors discussed in

this paper, considerations such as logistical feasibility and product accessibility should also be taken into account in the future policy formulation for nirsevimab administration in China.

Our study has several limitations. First, as a hospital-based birth cohort, the inclusion of cases and follow-up relied on patients seeking care at the children's hospital. The omission of ALRI visits or hospitalizations at hospitals outside of SCH could lead to an underestimation of the incidence density of RSV. The analysis in this study only covers the cohort population residing in Suzhou city, which partially addresses this limitation. Meanwhile, if children who would seek treatment at SCH after falling ill have never visited SCH due to consistently staying healthy, it may result in an underestimation of the denominator in this study, consequently leading to an overestimation of hospitalization rates. Nevertheless, considering the low probability of children consistently remaining healthy without falling ill before the age of 5, the chances of overestimating hospitalization rates is also minimal. Second, the RSV hospitalization rate utilized in this study's model evaluation spans from 2016 to 2020, primarily predating the COVID-19 pandemic. Several researches revealed that the seasonal characteristics of RSV have changed in the post-COVID era due to the impact of the pandemic [26–28]. These alterations in seasonal trends are likely to influence the results of strategy evaluation. Hence, it is imperative to continue monitoring the RSV activity and observe whether the post-pandemic RSV seasonal trends gradually revert to the pre-pandemic winter-spring pattern or undergo permanent changes. Third, the incidence data for this study were obtained from the Suzhou region in southeast China. RSV activity trends can vary across different geographical regions, potentially resulting in variations in the optimal strategies. Therefore, caution is warranted when extrapolating the findings of this study to other areas. Despite these limitations, the estimated incidence density of RSV-ALRI in this study enriches the available data on population-based RSV disease burden in mainland China, providing a foundation for subsequent health economics evaluation. Additionally, the optimal mAb administration strategies identified in this study can provide evidential support for the formulation of the RSV monoclonal antibody administration policy in Suzhou, China.

In conclusion, this study found that administering monoclonal antibody to children born in July to January at birth and administering to the remaining <1 years old children in September or October would be the optimal seasonal passive immunization strategy for children in Suzhou, China.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Genming Zhao reports financial support was provided by Merck Sharp & Dohme Cor.

Data availability

Data will be made available on request.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2023.11.054>.

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